Venkata Lakshmi Parimala Pasupuleti

Agentic AI Engineer · LangGraph · Claude Code · RAG · Tool Calling · LLM Evaluation & Guardrails · Python · AWS · Production AI Systems

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github.com/venkatalakshmiparimala/AgentProbe | venkatalakshmiparimala.github.io/Portfolio | Virginia, USA (Gaithersburg MD area)

PROFESSIONAL SUMMARY

Agentic AI engineer with 8 years of production software experience and hands-on projects that fit the AstraZeneca bar exactly: **something personally built and shipped that people use**. Built **Leasel** — a production **8-agent LangGraph system** with tool calling, durable state, RAG, citation grounding, evaluation harness, and **HITL guardrails** enforcing human oversight at every regulated decision point. Built **AgentProbe** — an open-source **LLM evaluation and guardrails toolkit** with 5-metric eval engine, LLM-as-judge scoring, CI regression harness, and multi-provider routing across **Claude (Anthropic), OpenAI, Groq, and Ollama**. Use **Claude Code daily** as the primary development environment — not occasionally, but as core workflow. At Magizhs: deployed an **MCP server** integrating agentic workflow outputs into downstream operational tooling. At Entrata (8yr): shipped production systems in a **regulated, data-sensitive environment** with 100% PHI compliance — the same standard a life-sciences platform requires. At GMU: explained complex technical systems to 600+ students — the communication pattern this role needs for scientists. Python-primary; TypeScript, Java; REST APIs; AWS; Docker, Kubernetes; GitHub Actions CI/CD. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Agentic AI Systems (LangGraph / LangChain) | Multi-Agent Orchestration | Tool Calling & Function Calling | MCP (Model Context Protocol) | RAG & Retrieval Pipelines | Vector Databases (ChromaDB / Qdrant) | Prompt Engineering & Structured Outputs | LLM Evaluation & Guardrails | LLM-as-Judge Scoring | HITL / Human Oversight Patterns | Claude Code (daily) | Claude / Anthropic API | OpenAI / Azure OpenAI | Multi-Provider LLM Routing | Python / FastAPI | TypeScript / JavaScript | Java / Spring Boot | REST APIs & SQL | AWS (S3 / Lambda / Bedrock / EC2) | Docker / Kubernetes | CI/CD (GitHub Actions) | Regulated / PHI-Compliant Environments | HIPAA / Data Security

EXPERIENCE

Blue Horse Digital

Sep 2026 - Present

AI/ML Engineer (Volunteer) — Virginia, USA

- Building AI-enabled applications and data pipelines using **Claude Code** as primary development environment; integrating agentic components with backend systems; surfacing insights through application-facing APIs.

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built an **agentic workflow system** (LangGraph stateful graph: parse → plan → retrieve → validate → route) with **RAG over a regulatory corpus**, citation-grounded compliance checks, validation guardrails, and **MCP server deployment** connecting agent outputs to downstream enterprise tooling.

Entrata — PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer — Lehi, UT (Remote)

- **Regulated, data-sensitive environment** — 100% PHI compliance, Libsodium-encrypted pipelines, multi-tenant data isolation at the schema layer; zero security incidents across the full AWS S3 migration. The accountability standard life-sciences platforms require.
- Shipped 20+ production modules (Python, Java, PHP, PostgreSQL, RabbitMQ) across 120+ Agile sprints; owned compliance pipelines, async processing systems, and document management infrastructure at 99.5% uptime for 100K+ monthly active users.
- **Explained technical decisions to non-technical stakeholders** across 300+ code reviews and cross-team incident responses; mentored 5+ engineers; built debugging tools reducing resolution time by 40%.

Entrata India — Db Xento Systems

Jan 2017 - Sep 2018

Associate / Trainee Software Engineer — Pune, India

- Built compliance modules (PHP, Python, Java, PostgreSQL); led AWS S3 cloud migration; income-limits import (1.6M HUD/Tax Credit records across 5 years).

PROJECTS

Leasel: Production Agentic AI System — Built, Shipped, Used

LangGraph · Tool Calling · RAG · Evaluation · HITL Guardrails

Production agentic AI system personally built and deployed: **8-agent LangGraph StateGraph** with durable state, **tool calling** across heterogeneous data sources, and **multi-step reasoning**. Agents: Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → **Human Oversight gate** (fires when AI confidence is insufficient). RAG layer: hybrid TF-IDF + semantic reranking, **citation grounding** to exact source text, missing/conflicting clause detection. Core safety rule enforced in code: LLM extracts and reasons; a human or deterministic logic makes every regulated decision. Built with **Claude Code** as primary development environment. **Evaluation harness**: 5-metric engine, LLM-as-judge scoring, CI regression gate, offline + online eval loops — 93.6% precision, 87.9% F1 on 1,240 test cases.

Python · LangGraph · FastAPI · MCP · ChromaDB · OCR · SQLite · Claude Code · GitHub Actions · 2026

AgentProbe: Open-Source LLM Evaluation & Guardrails Toolkit

Claude · OpenAI · Groq · Ollama · Eval Harness · CI/CD

Open-source toolkit making LLM applications **trustworthy and observable**: **5-metric eval engine** (correctness, groundedness, latency, token efficiency, hallucination rate), **LLM-as-judge scoring**, CI-integrated regression harness with golden datasets, offline + online evaluation loops. **Multi-provider routing** across Claude (Anthropic), OpenAI, Azure OpenAI, Groq, and Ollama — the same provider stack AZ uses. Prompt version management; reusable guardrails and evaluation patterns applicable to scientific AI contexts. Built and maintained using **Claude Code** daily. Live: github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · Claude API · OpenAI API · CI/CD (GitHub Actions) · 2025

AI on FHIR: NLP-Driven Scientific Data Integration

LLM Integration · REST APIs · Regulated Data · HIPAA

Platform connecting LLMs to a structured scientific data API (FHIR): translated natural-language queries into **structured API calls** using LLMs, BERT, and embeddings; enforced **HIPAA-compliant RBAC**; ~60% reduction in manual retrieval time. Demonstrates the core pattern this role requires: **LLM as interface to internal data and models**, with compliance constraints enforced in the integration layer, not assumed by the model.

Python · Flask · React · LLMs · BERT · FHIR APIs · Docker · spaCy · 2024

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia May 2026 | GPA 3.87

B.Tech Computer Science & Engineering — JNTU, Anantapur 2017